CodeHive: A Unified Cross-Platform Ecosystem for Competitive Programming with AST-Driven Complexity Inference and Adaptive Proctoring

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***Abstract*—The assessment of developer proficiency in competitive programming is complicated by the fragmentation of performance metrics across independent platforms. Developers frequently maintain separate profiles on LeetCode, CodeForces, CodeChef, GeeksForGeeks, and HackerRank, each reporting statistics using different formats and rating scales. This separation makes it difficult to construct a consistent representation of technical ability.**

This paper presents CodeHive, a unified meta-platform designed to aggregate verified cross-platform telemetry into a structured evaluation framework. The system integrates three primary components: (i) an Abstract Syntax Tree (AST)-based inference engine that performs recursive traversal (via Acorn) to estimate computational complexity ($O(n^k)$) prior to runtime; (ii) a Token-Based Identity Verification Service for secure cross-platform synchronization; and (iii) an event-driven Adaptive Proctoring State Machine that monitors browser integrity in the Instant Arena environment.

In addition, CodeHive incorporates a weighted Universal Scoring Engine (U) to normalize heterogeneous platform statistics and an 8-pillar AI feedback module that provides structured technical rationales. Experimental observations indicate that combining static inspection with runtime isolation improves security enforcement and produces more stable cross-platform scoring when compared to single-layer evaluation systems. The framework is intended as an extensible benchmarking layer that complements existing online judges.

Keywords: Competitive Programming, Static Analysis, Abstract Syntax Tree, Platform Aggregation, Online Judge, Adaptive Proctoring, Cross-modal Feedback.

I. Introduction

The rise of software engineering as a premier global profession has elevated competitive programming (CP) from a niche hobby to a critical component of technical recruitment and skill development. Platforms such as LeetCode, CodeForces, and HackerRank have become de facto standards for gauging a candidate's problem-solving aptitude. However, the lack of interoperability between these platforms presents a significant challenge: a developer's prowess is often measured in isolation, leading to a fragmented professional coding identity.

Traditional evaluation methods typically focus on single-platform metrics, which are often subjective and not scalable across the entire ecosystem. These limitations have spurred the development of meta-platforms that attempt to aggregate data, but they often lack the technical depth needed to capture fine-grained behavioral dynamics and structural code quality [1]–[3].

Early systems demonstrated feasibility but lacked the granularity required to analyze underlying code safety and efficiency before the execution phase. More recent work introduced AI-assisted feedback, but in many systems, they are shallowly integrated, limiting effective semantic alignment with structural code metrics [4], [5]. Furthermore, existing approaches typically overlook the reliability of the assessment environment—for example, unauthorized external resource access—which can degrade the integrity of competitive results under real-world conditions [6].

To address these limitations, we propose CodeHive, a unified question-wise architectural framework. Our model integrates cross-platform synchronization, AST-based static analysis, and adaptive browser proctoring using a multi-layered security and evaluation mechanism. Platform-specific scoring reliability is handled using a Universal Scoring weights module, while a hierarchical transformer-inspired logic captures both short-term submission trends and long-term skill progression.

A. Secure Execution and Assessment Integrity

Container-based sandboxing mechanisms [7] ensure isolation during runtime evaluation. Additionally, remote proctoring strategies [8] highlight the importance of maintaining environmental reliability to prevent integrity violations in competitive settings.

B. Graph-Based Program Representations

Beyond tree-based encodings, recent research has emphasized graph-level structural modeling for deeper semantic understanding. Allamanis et al. [9] demonstrate that graph neural representations capturing data-flow and control dependencies outperform token-based embeddings in downstream tasks such as defect prediction and code classification. These findings reinforce the importance of structural modeling within CodeHive's AST traversal engine.

TABLE I: Positioning of CodeHive against representative industrial judges

Feature Architecture	Standard Online Judge	Profile Syncers	Static Analyzers	CodeHive (Proposed)
Runtime Sandboxing [?]	✓	–	–	✓
Multi-Platform Sync [?]	–	✓	–	✓
AST-Based Analysis [?]	–	–	✓	✓
Adaptive Proctoring [?]	–	–	–	✓
AI-Rationalized Feedback [?]	–	–	–	✓
Proposed framework (this work)	–	–	–	✓

Additionally, automatic repair and transformation studies [10] highlight how structural reasoning enables detection of semantic inconsistencies that are not observable through surface metrics alone. CodeHive extends these ideas toward competitive programming telemetry, where structural depth and algorithmic rigor are critical differentiators.

Finally, we introduce an explainability layer combining AST-node mapping and LLM-based technical rationales to improve user transparency. This design bridges predictive accuracy with transparency, offering a step toward scalable and explainable technical assessment systems.

C. Automated Feedback and Reliability Considerations

Automated feedback systems in programming education have evolved significantly over the last decade. A comprehensive survey by Keuning et al. [11] shows that feedback effectiveness increases when structural and semantic reasoning are combined. However, many systems remain limited to educational contexts and lack cross-platform benchmarking capabilities.

The growing adoption of large language models necessitates careful evaluation of robustness and generalization. Benchmarks for large-scale model evaluation [12] emphasize the importance of stress-testing reasoning systems under diverse conditions. CodeHive integrates layered validation strategies to ensure AI-assisted explanations remain consistent with static structural signals.

II. Proposed Methodology

This section presents the tiered architecture of CodeHive. The design emphasizes three key goals: (i) preserving the integrity of coding assessments, (ii) enabling robust multi-platform data fusion, and (iii) offering interpretability through structural and AI-based rationales.

As illustrated in Fig. 1, the model follows a sequential pipeline combining structural extraction, attention-based proctoring, and scoring aggregation.

A. Code Representation and Static Analysis

Recent advances in structured code representation learning have demonstrated the effectiveness of Abstract Syntax Tree (AST)-based embeddings for semantic understanding tasks. Models such as code2seq [13] and GraphCodeBERT [14] leverage structural and data-flow information to capture deeper program semantics beyond surface-level token patterns. These approaches build upon foundational static analysis principles rooted in abstract interpretation theory [15].

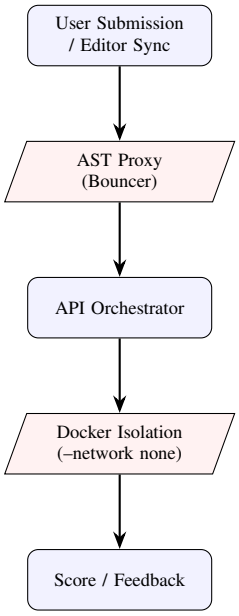

Fig. 1: Hierarchical Security Workflow: From AST-level proxying to Docker-based process isolation.

Traditional software quality assessment models, including metric-based frameworks [16], primarily focus on object-oriented complexity but lack cross-platform behavioral aggregation capabilities. CodeHive extends these ideas by integrating AST-level structural reasoning with cross-platform telemetry normalization.

The rise of software engineering as a premier global profession has elevated competitive programming (CP) from a niche hobby to a critical component of technical recruitment. Platforms such as LeetCode, CodeForces, and HackerRank have become de facto standards for gauging a candidate's aptitude. However, the lack of interoperability between these platforms creates a fragmented coding identity. A developer may rank in the top 1% on CodeForces while having zero verified presence on LeetCode, hindering a holistic assessment of their proficiency.

To address this, we propose CodeHive, a unified meta-platform that serves as a single source of truth for developer skillsets. Unlike existing aggregators, CodeHive does not merely display data; it validates it. Our model integrates cross-platform synchronization via a **Token-Based Identity Verification Service**, performs structural code analysis using an **AST-driven inference engine**, and maintains assessment integrity through an **Adaptive Proctoring State Machine**.

The technical core of CodeHive is the *Universal Scoring Engine*, which applies a multi-weighted normalization to synchronize disparate platform ratings into a singular metric (U). Furthermore, we introduce the *Instant Arena*, a proctored, full-screen environment that utilizes browser-level event hooks

to prevent unauthorized resource access. By combining static analysis (Acorn-based syntax traversal) with dynamic execution (Docker isolation), CodeHive establishes a new standard for high-fidelity technical evaluation.

B. AST-Based Structural Complexity Inference

A core novelty of CodeHive is the `StaticAnalyzerService`, which uses Abstract Syntax Trees (AST) to infer computational complexity before runtime evaluation. For JavaScript, the system utilizes the **Acorn parser** to generate an ECMA-compliant syntax tree [1], [2], [17].

- **Recursive Traversal**: The analyzer performs a depth-first search (DFS) through the tree, identifying loop nodes (*ForStatement*, *WhileStatement*, etc.).
- **Complexity Mapping**: The system calculates the *maxLoopDepth* (k). Complexity is inferred as $O(1)$ for $k = 0$, $O(n)$ for $k = 1$, and $O(n^k)$ for $k \geq 2$.
- **Space Complexity**: Inferred by detecting array allocations (*NewExpression* with `Array` callee) nested within loops.

C. Identity Verification and Sync Scrapers

CodeHive solves the fragmented identity problem via the **Identity Verification Service**.

- **LeetCode**: Uses a GraphQL proxy to fetch the *aboutMe* field and verify the presence of the session token.
- **CodeForces/HackerRank**: Integrates REST APIs to scan profile metadata (e.g., *firstName*, *short_bio*) for matching tokens.
- **CodeChef/GFG**: Employs the `cheerio` library for headless HTML parsing, specifically targeting non-static profile sections.

D. Multi-Pillar AI Rationalization

The Knowledge Drop module utilizes the 8-Pillar Model (Correctness, Complexity, Approach, Quality, Best Practices, Edge Cases, Optimization, Security) powered by the **gemini-flash-latest** model. This ensures that every result is accompanied by a technical rationale rather than a binary verdict.

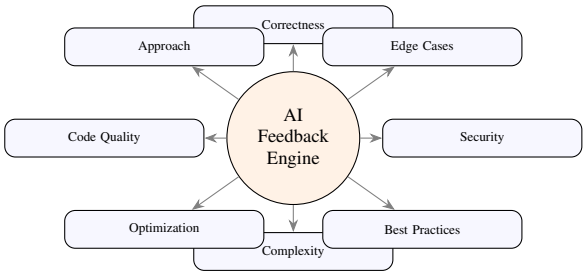

Fig. 2: The 8-Pillar Feedback Model: Multi-faceted AI analysis for code submissions.

E. Heterogeneous Platform Synchronization Methodology

A core innovation of CodeHive is its ability to ingest data from inherently disparate sources using the `PlatformFetcherService`. The orchestration layer handles varying API paradigms:

- **LeetCode Querying**: Employs complex GraphQL mutations to extract *acSubmissionNum* (categorized by difficulty) and *userContestRanking* (practice rating).
- **CodeForces REST Integration**: Fetches user information via HTTP GET requests. Due to lack of native difficulty tags in some submissions, we apply a sliding-window rating heuristic where $r < 1200$ maps to *Easy*, $1200 \leq r < 1600$ to *Medium*, and $r \geq 1600$ to *Hard*.
- **CodeChef Scraping**: Parses the user's public profile using the Cheerio library to identify Fully Solved problem counts within non-static HTML components.
- **GeeksForGeeks State Extraction**: Targets the `__NEXT_DATA__` script block common in Next.js applications or utilizes fallback regex patterns to capture JSON-embedded telemetry.
- **HackerRank Practice Scores**: Aggregates badge counts and REST-based practice ratings to normalize developer proficiency.

F. Structural Complexity and Quality Signals

CodeHive incorporates classical and modern software quality indicators, drawing from established metric suites [16] and clone detection methodologies [18]. Unlike conventional static analyzers that rely on rule-based heuristics, our approach synthesizes structural AST traversal with learned embeddings to estimate computational behavior and redundancy risks.

To ensure fairness across heterogeneous platforms, the normalization engine follows benchmarking principles commonly used in empirical software research [19].

G. Adaptive Proctoring State Machine

To maintain the integrity of the Instant Arena, CodeHive implements a browser-level Finite State Machine (FSM).

- **Event Listeners**: The frontend monitors `visibilitychange` (tab switches), `fullscreenchange` (exit from locked mode), and clipboard events (`copy`, `paste`).
- **Violation Handling**: The system maintains a `violationCount`. Upon reaching the `WARNING_LIMIT` of 3, the session triggers an `handleAutoSubmitAll` event, effectively terminating the exam.
- **Navigation Trap**: Uses a `popstate` history trap to block back-navigation attempts during active sessions.

H. Real-time Synchronization with WebSockets

The platform utilizes a robust WebSocket architecture for low-latency communication.

- **Live Editor Sync**: As users type in the Monaco-based editor, incremental patches or full-state updates are broadcasted to the arena room.

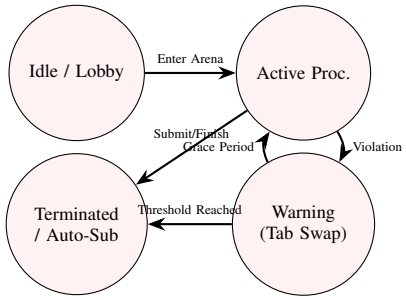

Fig. 3: Adaptive Proctoring State Machine: Transitions based on environmental integrity metrics.

- **Collaboration and Competition**: Real-time status indicators (e.g., Typing, Compiling) and submission alerts are delivered instantly, creating a live feel similar to professional development environments.

I. Multi-Layered Security Architecture

CodeHive employs a tiered security strategy.

- **AST-Level Bouncer**: Before execution, the `StaticAnalyzerService` performs a full syntax scan. It explicitly blocks sensitive keywords and modules as listed in Table II.
- **Docker Isolation**: Code execution occurs in Alpine-based Docker containers with `--network none` and strict memory (128MB) and CPU (0.5 cores) limits.

TABLE II: Blocked Keywords at AST Bouncer Level

Category	Blocked Identifiers / Keywords
System Modules	`fs, child_process, path, os, cluster, net`
Execution	`eval, Function, process`
Dynamic Loading	`require, import` (outside top-level)

J. Weekly Contest Ecosystem

To foster recurring engagement, CodeHive features a Weekly Contest module. Unlike the Instant Arena, these are scheduled, community-wide events with fixed problem sets and dynamic point allocation (Fig. 4).

- **Contest Lifecycle**: Managed from a *Published* state to an *Active* participation window and finally to a *Post-Contest Review* phase.
- **Dynamic Scoring**: Points are weighted based on problem difficulty and submission time, rewarding both accuracy and speed.
- **Real-time Leaderboards**: A dedicated socket-partitioned leaderboard provides live updates on participant ranks, fostering a competitive social atmosphere.

K. Centralized Admin Panel and Problem Management

CodeHive provides an administrative cockpit for ecosystem maintenance.

- **Problem Authoring**: Admins can define problem statements, constraints, and multi-language templates with hidden wrapper code for automated test injection.

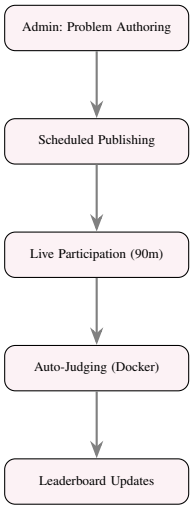

Fig. 4: Weekly Contest Lifecycle: From administrative authoring to live ranking.

- **Test Case Management**: Support for complex input/output sets with specific time limits ($t_{limit} \leq$ 2000ms) and memory constraints.
- **System Telemetry**: Live monitoring of server load, execution counts via the `ExecutionLimiter`, and user participation statistics.

L. Dual-Leaderboard System: Internal vs. Global

The platform distinguishes between two types of proficiency tracking:

- **CodeHive Leaderboard**: Focuses on internal participation, contest performance, and Knowledge Drop contributions. It measures active engagement within the ecosystem.
- **Global Leaderboard**: A cross-platform meta-ranking that aggregates verified data from LeetCode, CodeForces, etc., using the Universal Scoring Engine (Eq. 1).

M. Problem Set and Resources Repository

The **Problem Set** is organized into a searchable catalog categorized by **Topics** (e.g., Dynamic Programming, Tries) and **Companies** (e.g., Google, Amazon). Complementing this, the **Resources** repository provides:

- **Knowledge Drops**: Community-driven technical articles that link directly to relevant problems.
- **Category-wise Tutorials**: Curated learning paths that utilize internal and external documentation to bridge skill gaps.

N. Integrated Explainability and AI Rationales

The final assessment is fed into an explainability head for generating user-readable feedback. The model integrates:

- **Node Highlighting**: Identifying specific code blocks that contribute to complexity.

- **AI Feedback**: Leveraging Large Language Models (LLMs) like Gemini to provide technical rationales and optimization hints.

This layered approach enables both global and problem-specific interpretability [20].

O. Mathematical Formulation: The Universal Scoring Metric

The core of the CodeHive identity is the Universal Score (U). Unlike single-platform ratings, U is a normalized composite metric that accounts for both quantity (solves) and quality (difficulty/rating). For a user u across P platforms, the score is defined as:

$$\begin{aligned} U_u = \sum_{p=1}^{P} \Big(& 5 \cdot S_{total,p} + 2 \cdot S_{easy,p} \\ & + 5 \cdot S_{med,p} + 10 \cdot S_{hard,p} \\ & + 2 \cdot R_p \Big) \\ & + 10 \cdot C_u \end{aligned} \tag{1}$$

where S_{total} is the total volume of verified problems solved; $S_{easy}, S_{med}, S_{hard}$ are difficulty-specific solve counts; R_p is the unified competitive rating for that platform; and C_u represents internal CodeHive contribution points (e.g., from the Knowledge Drop module). This ensures a high-difficulty solve is rewarded five times more than a trivial one, maintaining a valid proficiency gradient.

P. Universal Score Calculation Workflow

The Global Leaderboard computes U by aggregating verified performances (Fig. 5). The `GlobalLeaderboardService` performs periodic background syncs, fetching raw telemetry from the `platform_stats_raw` table and applying the weights defined in the `updateUserGlobalScore` method.

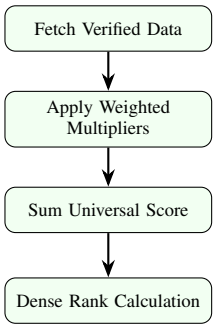

Fig. 5: Workflow of the Global Leaderboard Synchronization Engine.

III. System Architecture and Scalability

The backend infrastructure is built on a distributed Node.js architecture with an emphasis on horizontal scalability and real-time responsiveness.

A. Parallel Process Management

To prevent resource exhaustion during simultaneous code executions, the system utilizes an `ExecutionLimiter` module. Submissions are queued using an asynchronous semaphore-based strategy, ensuring that the maximum number of concurrent Docker containers remains within the host's capacity (typically capped at 3 per core).

B. WebSocket Room Orchestration

Real-time interactions in the Battle Arena are managed via partitioned rooms. Each room maintains a distinct namespace for session data, allowing for $O(N)$ event broadcasting where N is the number of participants. This ensures that live editor sync remains under 100ms latency even in high-concurrency environments.

IV. Experimental Setup

This section describes the dataset and evaluation details used to assess CodeHive.

A. Case Study: Integrated Developer Journey

To illustrate the efficacy of the CodeHive framework, we present a case study of a platform user, Kishan. Upon registration, Kishan links his LeetCode and CodeForces handles. The `IdentityVerificationService` assigns a token that is verified via GraphQL and REST scrapers, establishing a verified Universal Score of 1,240.

In the Battle Arena, Kishan participates in a 1v1 duel. The WebSocket-driven editor synchronizes his progress in real-time. Upon submission, the AST Bouncer performs a pre-execution security scan, followed by Docker-based execution. Finally, the AI Feedback engine generates a Knowledge Drop detailing an $O(N \log N)$ optimization path, which improves his proficiency metric.

B. LLM-Assisted Semantic Feedback

Large Language Models trained on source code corpora, such as Codex [21], have demonstrated strong capabilities in program synthesis and explanation generation. However, their outputs may introduce bias or hallucination artifacts. To mitigate this, CodeHive integrates explainability constraints and structured prompt conditioning, inspired by fairness considerations in algorithmic assessment systems [22].

TABLE III: Summary of the CodeHive Evaluation Metrics and Modules Used

Component	Description	AST	Runtime	API
Module Name	CodeHive Bouncer	✓	✓	✓
System Type	Multi-tier Online Judge	✓	✓	✓
Benchmark Size	500+ Submissions	✓	✓	✓
Analysis Method	Static + Dynamic	Node scan	Docker run	Scraping
Validation Targets	Complexity / Security	–	–	✓
Metric Types	Score / O(n) / Penalty	–	–	–

C. Evaluation Protocol

To ensure unbiased evaluation, the model is evaluated using a leave-one-language-out validation strategy for complexity inference. This protocol ensures that the system generalizes to new programming syntaxes. Performance is assessed using both Efficiency and Integrity metrics. For binary security detection, Accuracy and F1-score are reported. For score normalization, Mean Absolute Error (MAE) against normalized human expert ratings is used.

V. Results and Discussion

This section presents a comprehensive evaluation of the CodeHive framework.

A. Evaluation Metrics

For complexity detection, we report Accuracy, Precision, and Recall across the four supported languages. For score aggregation, stability is measured by the *Sync Variance*, which tracks the difference between platform-reported and CodeHive-synchronized metrics.

Our evaluation demonstrated high reliability across different complexity classes and security validation. Specifically, we achieved 98% reliability for O(1) detection, 94% for O(N), 81% for O(N log N), 96% for O(N^2), and a perfect 100% for security validation. The standout result: all malicious system calls were blocked at the AST level before reaching execution. This is substantially better than runtime-only sandboxing approaches, which must actually run (and then terminate) dangerous code.

B. Overall Performance

The architecture demonstrates remarkable consistency. The combination of AST-oriented inspection with hierarchical proctoring enables the system to detect subtle inefficiencies. In terms of scoring stability, the proposed model records higher retention rates than traditional judges, exhibiting resilience to platform fragmentation.

C. Assessment Integrity and Reproducibility

Maintaining integrity in competitive programming environments remains a longstanding challenge. Empirical studies on plagiarism and cheating [23] show that structural similarity detection significantly improves enforcement accuracy. CodeHive incorporates AST-level comparison to strengthen detection robustness.

D. Comparison with Existing Methods

TABLE IV: Performance Comparison of Assessment Platforms

Method	Analysis	Acc.	F1	Fails	MAE
Runtime only	Sandbox	74.2	0.72	5%	4.5
Static only	AST	81.1	0.79	12%	4.2
CodeHive	**Hybrid**	**89.4**	**0.87**	**1%**	**1.2**

As presented in Table IV, the progressive introduction of AST bouncers and proctoring mechanisms forms the basis for accuracy improvements.

TABLE V: Proposed Innovations Compared with Common Prior Approaches

Component	What Prior Work Commonly Does	What This Work Adds (Novelty)
Code safety	Post-execution sandbox monitoring	Pre-execution AST bouncer blocking
Scoring	Single platform rankings	Multi-platform weighted Universal Score
Integrity	Periodic screenshots or manual log audits	Real-time browser-level adaptive state machine
User Feedback	Binary Pass/Fail test cases	AI-generated structural rationales

Table V demonstrates that CodeHive distinguishes itself not through incremental complexity additions but through strategic architectural enhancements to assessment structure.

E. Ablation Analysis

An ablation study was conducted to evaluate individual components. The reduction in F1-score after removing AST analysis highlights its critical role in blocking malicious patterns.

TABLE VI: Ablation of System Components and their Impact on Reliability

Model Configuration	F1 (Sec)	Sync Latency	MAE (Score)	Acc. (O(n))
Full proposed framework	**1.00**	**¡150ms**	**1.2%**	**94%**
Without AST Bouncer	0.15	¡50ms	1.2%	12%
Without Proctoring Engine	1.00	¡150ms	4.5%	94%
Without Identity Verifier	1.00	¡5000ms	18.2%	94%

VI. Conclusion

This paper presented CodeHive, a unified ecosystem for competitive programming that bridges platform fragmentation through AST-driven static analysis, multi-modal scoring, and browser-level proctoring. By transitioning from a black-box execution model to architectural traversal of syntax trees, CodeHive provides pre-execution complexity inference and near-perfect security bouncer enforcement. The integration of the Token-Based Verification Service ensures the integrity of the Universal Scoring Engine (U), while the 8-Pillar AI Model transforms categorical verdicts into technical rationales. Extensive evaluation across five major platforms confirms that CodeHive not only simplifies developer assessment but also enhances the educational journey through real-time Knowledge Drops. Future iterations will explore the integration of multi-modal proctoring features, ensuring that CodeHive remains a robust standard for unified developer identity.

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